

How should dehydration be assessed and managed in an adult, including fluid selection for mild, moderate, and severe cases, special considerations for hypernatremic dehydration, hypovolemic shock, and comorbid cardiac or renal disease?

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Assessment and Management of Dehydration in Adults

Assessment of Dehydration

Directly measured serum or plasma osmolality >300 mOsm/kg is the gold standard for diagnosing low-intake dehydration in adults, particularly in older patients where clinical judgment and renal parameters are unreliable. 12

Diagnostic Approach

- **Serum osmolality >300 mOsm/kg** confirms low-intake dehydration when glucose and urea are within normal ranges **1.2**
- When direct osmolality measurement is unavailable, use the **calculated osmolality equation**: $1.86 \times (\text{Na}^+ + \text{K}^+) + 1.15 \times \text{glucose} + \text{urea} + 14$ (all in mmol/L) with action threshold >295 mmol/L **2**
- **Distinguish between low-intake dehydration** (hypernatremic, raised osmolality) and **volume depletion** (from vomiting/diarrhea, normal or low osmolality) as they require different management **1.2.3**

Clinical Signs to Assess Severity

- **Mild-to-moderate dehydration**: Tachycardia, postural pulse changes ≥ 30 bpm, postural dizziness, dry mucous membranes, decreased skin turgor **4**
- **Severe dehydration**: Altered mental status, hypotension, oliguria (< 0.5 mL/kg/hour), rapid breathing, hypovolemic shock **5.4**
- **Monitor weight changes and serial vital signs** throughout therapy to assess adequacy of rehydration **5**

Management by Severity

Mild-to-Moderate Dehydration

Reduced osmolality oral rehydration solution (ORS) containing 65–70 mEq/L sodium and 75–90 mmol/L glucose is first-line therapy for mild-to-moderate dehydration in adults. **6**

- **Oral rehydration**: Give ORS as much as the patient wants; for structured dosing, adults should receive as much as desired after each loose stool **5.6**
- Supplement with **locally available fluids** (cereal-based gruels, soup, rice water) but **avoid soft drinks** due to high osmolality **5**
- **Nasogastric ORS** may be used in patients with moderate dehydration who cannot tolerate oral intake, with total fluid prescription 2200–4000 mL/day depending on ongoing losses **6**
- **Subcutaneous rehydration** (hypodermoclysis) is an **acceptable alternative when IV access is difficult** in stable elderly patients, using isotonic or hypotonic solutions at rates up to 3000 mL over 24 hours **4.7**

Severe Dehydration

Isotonic crystalloids (0.9% normal saline or balanced salt solutions like lactated Ringer's) are the **fluids of choice for severe dehydration**, with an initial bolus of 15–

20 mL/kg/hour (1–1.5 L in average adults) during the first hour in the absence of cardiac compromise. [6.4](#)

- For sepsis with hemodynamic instability, give 20 mL/kg bolus initially and continue rapid administration until clinical signs improve (hypotension, low urine output, impaired mental status resolve) [6](#)
- After initial resuscitation, continue IV fluids at 4–14 mL/kg/hour until urine output exceeds 0.5 mL/kg/hour and perfusion markers normalize [6.4](#)
- Fluid administration rate must exceed ongoing losses: urine output + 30–50 mL/hour insensible losses + gastrointestinal losses [6](#)

Special Considerations

Hypernatremic Dehydration

In hypernatremic dehydration, sodium correction must not exceed 8 mEq/L in 24 hours or 0.5 mEq/L/hour to prevent osmotic demyelination syndrome and cerebral edema. [4.8.9](#)

- Calculate corrected sodium: Add 1.6 mEq/L to measured sodium for every 100 mg/dL glucose above 100 [4](#)
- If corrected sodium is normal or elevated, use 0.45% saline at 4–14 mL/kg/hour [6.4](#)
- For hyperosmolar dehydration in elderly, use hypotonic solutions (half-normal saline + 5% glucose or 5% dextrose) to replace deficit while diluting elevated osmolality [7](#)
- Avoid initial fluid boluses in hypernatremic dehydration; rapid rehydration rate is the most significant risk factor for cerebral edema [8](#)
- Safe rehydration rate is <6.8 mL/kg/hour based on ROC curve analysis [8](#)
- Monitor serum sodium every 2–4 hours initially during active resuscitation, then every 24–48 hours once stable [4](#)

Hypovolemic Shock

For hypovolemic shock, immediately initiate isotonic crystalloids at 15–20 mL/kg/hour with continuous hemodynamic monitoring until shock resolves. [6.4](#)

- Continue aggressive resuscitation until urine output >0.5 mL/kg/hour, heart rate normalizes, blood pressure increases ≥10%, mental status improves, and peripheral perfusion restores (warm extremities, capillary refill <2 seconds) [4](#)

- Serial lactate levels should trend downward and base deficit should normalize with successful fluid therapy **4**
- Reassess after each 250–500 mL bolus to determine further fluid needs **4**

Cardiac or Renal Disease

In elderly patients with cardiac or renal disease, start with 500–1000 mL crystalloid over 30 minutes to 1 hour as the initial bolus, not the standard 1–1.5 L bolus. **2**

- Monitor closely for fluid overload signs: increased jugular venous pressure, increasing crackles/rales, peripheral edema **2**
- **Reduce infusion rate immediately** if signs of fluid overload develop **2**
- Elderly patients mobilize extracellular water more slowly, particularly during inflammatory processes, necessitating fluid restriction **2**
- **Never reflexively give IV fluids based solely on "NPO for 3 days" without assessing actual volume status**—this can precipitate acute respiratory failure requiring intubation in patients with occult heart failure **2**
- Consider **subcutaneous fluid administration** in stable elderly patients as it reduces agitation (37% vs 80% with IV), lowers complication rates, and decreases infection risk **2**

Fluid Selection Algorithm

Step 1: Assess Hemodynamic Stability

- **Hemodynamically unstable** (shock, severe hypotension): Use isotonic crystalloids (0.9% normal saline or lactated Ringer's) **6,4,7**
- **Hemodynamically stable**: Proceed to Step 2 **2**

Step 2: Calculate Corrected Sodium

- Corrected sodium = measured sodium + $1.6 \times [(\text{glucose} - 100) / 100]$ **4**

Step 3: Select Fluid Based on Corrected Sodium

- **Low corrected sodium**: Use 0.9% normal saline at 4–14 mL/kg/hour **6,4**
- **Normal or elevated corrected sodium**: Use 0.45% saline at 4–14 mL/kg/hour **6,4**
- **Hyperosmolar dehydration in elderly**: Use hypotonic solutions (half-normal saline + 5% glucose or 5% dextrose) **2**

Step 4: Adjust for Comorbidities

- **Cardiac or renal disease:** Reduce initial bolus to 500–1000 mL over 30–60 minutes and monitor closely for overload ⁷
- **Hypernatremia:** Avoid boluses; maintain rehydration rate <6.8 mL/kg/hour ⁸

Critical Pitfalls to Avoid

- **Do not give rapid fluid boluses in hypernatremic dehydration**—this is the most significant risk factor for cerebral edema ⁸
- **Do not administer aggressive fluid boluses in elderly patients with renal impairment** to prevent acute pulmonary edema ⁶
- **Do not withhold fluids solely because BUN/creatinine are low** when signs of hypovolemia persist (urine output <0.5 mL/kg/hour, tachycardia, hypotension)—this leads to continued tissue hypoperfusion ⁴
- **Do not persist with aggressive fluids when urine output exceeds 0.5 mL/kg/hour** and clinical perfusion markers have normalized—this increases risk of iatrogenic fluid overload ⁴
- **Do not correct sodium faster than 8 mEq/L in 24 hours** to avoid osmotic demyelination syndrome ^{4,9}
- **Do not use soft drinks for rehydration** due to high osmolality ⁵
- **Dehydrated adults who fail to respond promptly to ORS** should be reassessed to exclude cholera or other severe pathology ⁵

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